Canvas Quiz 2, Modules 2–3 (graded)

Module 2 - IV Bolus Administration · Canvas quiz · No date

Name _____ Date _____

Question 1 Which of the following best describes the volume of distribution?

A. hypothetical volume of body fluid that would be required to dissolve the total amount of drug at the same concentration as that found in the blood
B. proportionality constant relating the rate of drug absorption to the rate of drug elimination
C. estimate of the time course of drug absorption, distribution, metabolism, and elimination
D. physiological measure of the volume occupied by the peripheral compartments

Question 2 A 250-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug. A = 10.16 mg/L, B = 5.65 mg/L, $\alpha = 3.59\ hr^{-1}$, $\beta = 0.15\ hr^{-1}$. What is the elimination half-life of this agent? Round to the nearest hundredth (two decimal places).

Question 3 A 250-mg IV bolus dose of a drug was administered to six healthy volunteers. The parameters below best describe the pharmacokinetics of the drug. A = 10.68 mg/L, B = 3.13 mg/L, $\alpha = 3.55\ hr^{-1}$, $\beta = 0.15\ hr^{-1}$. What is the concentration of the drug in the plasma 4 hours following administration of the dose? Round to the nearest hundredth (two decimal places).

Question 4 Why is a loading dose used?

A. To immediately obtain a therapeutic plasma concentration
B. To reduce the elimination half-life
C. To reduce the adverse effects associated with the drug
D. To improve patient adherence

Question 5 An antibacterial agent with a volume of distribution of 22 L and a clearance of 3.2 L/hr was infused intravenously until a concentration of 14 mg/L was reached. What was the concentration of drug in the plasma 5 hours following cessation of the infusion? Round to the nearest tenth (one decimal place).

Question 6 Which of the following results when the rate of infusion is increased?

A. Steady-state concentration decreases
B. The time to reach steady-state decreases
C. The time to reach steady-state increases
D. Steady-state concentration increases

Question 7 Recommend an infusion rate to achieve a steady-state concentration of 23 mcg/mL for an agent with an apparent volume of distribution of 15 L and half-life of 4 hr. Round to the nearest whole number.

Question 8 An analgesic drug is to be administered by continuous intravenous infusion at a rate of 38 mg/hr to a 60 year-old, 154 pound female patient. The drug has a half-life of 4 hours and an apparent volume of distribution of 0.21 L/kg. What is the expected steady-state concentration? Round to the nearest tenth (one decimal place).